

Lecture Scribe

CSE 400: Fundamentals of Probability in Computing

Tutorial-2

Source: Tutorial-2 Document

This scribe organizes and reconstructs the mathematical concepts, definitions, distributions, and general derivation techniques required to solve the problems in Tutorial-2. It is structured conceptually rather than question-wise and contains only material traceable to the provided document.

1. Continuous Uniform Distribution on an Interval

1.1 Definition

A continuous random variable X is Uniform on (a, b) if:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

The cumulative distribution function (CDF):

$$F_X(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x < b \\ 1, & x \geq b \end{cases}$$

The expectation:

$$E[X] = \frac{a+b}{2}$$

1.2 Random Point on a Line Segment

If a point is chosen at random on a line segment of length L , this means:

$$X \sim \text{Uniform}(0, L)$$

A geometric probability involving ratios (e.g., shorter/longer segment) requires:

- Expressing the ratio in terms of X
- Identifying regions of the sample space where the inequality holds
- Computing probability via integration over the appropriate interval

For continuous variables:

$$P(A) = \int_{\text{region}} f_X(x) dx$$

Since the density is constant for uniform distributions, probability reduces to:

$$\frac{\text{length of favorable region}}{\text{total length}}$$

2. Normal (Gaussian) Distribution and Bayesian Classification

2.1 Definition

A random variable $X \sim N(\mu, \sigma^2)$ has pdf:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Key parameters:

Mean: μ

Variance: σ^2

2.2 Mixture Models and Prior Probabilities

When a point is drawn from two possible populations:

With probability α , it comes from one distribution.

With probability $1 - \alpha$, from another.

This forms a mixture distribution:

$$f(x) = \alpha f_1(x) + (1 - \alpha) f_2(x)$$

2.3 Posterior Probability (Bayes' Rule)

If A and B partition the sample space:

$$P(A | x) = \frac{f_A(x)P(A)}{f_A(x)P(A) + f_B(x)P(B)}$$

Equal error probability condition implies balancing posterior probabilities.

Thus, one sets:

$$P(A \mid x) = P(B \mid x)$$

leading to an equation in prior probability α .

3. Minimization of Expected Absolute Deviation

We are interested in minimizing:

$$E[|X - a|]$$

3.1 General Form

$$E[|X - a|] = \int_{-\infty}^{\infty} |x - a| f_X(x) dx$$

Split at $x = a$:

$$= \int_{-\infty}^a (a - x) f_X(x) dx + \int_a^{\infty} (x - a) f_X(x) dx$$

Differentiating with respect to a gives:

$$\frac{d}{da} E[|X - a|] = P(X \leq a) - P(X \geq a)$$

Setting derivative to zero:

$$P(X \leq a) = \frac{1}{2}$$

Thus:

The minimizer of $E[|X - a|]$ is the median of X .

3.2 Uniform Case

For $X \sim \text{Uniform}(0, A)$:

$$\text{Median} = \frac{A}{2}$$

3.3 Exponential Case

If $X \sim \text{Exponential}(\lambda)$:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

The median m satisfies:

$$P(X \leq m) = \frac{1}{2}$$

$$1 - e^{-\lambda m} = \frac{1}{2}$$

This determines the optimal location.

4. Exponential Distribution and Memoryless Property

4.1 Definition

If $X \sim \text{Exponential}(\lambda)$:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$F_X(x) = 1 - e^{-\lambda x}$$

4.2 Memoryless Property

$$P(X > t + s \mid X > t) = P(X > s)$$

Proof:

$$\frac{P(X > t + s)}{P(X > t)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} = e^{-\lambda s}$$

4.3 Mixture of Exponentials

If with probability p_i , rate is λ_i ,
then survival probability:

$$P(X > t) = \sum_i p_i e^{-\lambda_i t}$$

Conditional survival:

$$P(X > t + s \mid X > t) = \frac{\sum_i p_i e^{-\lambda_i(t+s)}}{\sum_i p_i e^{-\lambda_i t}}$$

This is not memoryless unless rates are identical.

5. Poisson Distribution and Conditional Structure

5.1 Definition

$X \sim \text{Poisson}(\lambda)$

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Mean and variance:

$$E[X] = \lambda$$

5.2 Sum of Independent Poissons

If:

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

and independent, then:

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

5.3 Conditional Distribution Given Total

$$P(X = k \mid X + Y = n) = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Using independence:

$$= \frac{e^{-\lambda_1} \frac{\lambda_1^k}{k!} e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!}}{e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^n}{n!}}$$

This simplifies to a Binomial-type structure.

6. Transformation of Random Variables

6.1 CDF Method

If $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

6.2 Piecewise Transformations

If g is not one-to-one:

$$f_Y(y) = \sum_{x:g(x)=y} \frac{f_X(x)}{|g'(x)|}$$

Valid when inverse branches exist.

7. Binomial Approximation via Independent Movements

If each period:

Probability p of upward movement

Independent across periods

Then after n periods:

$$X \sim \text{Binomial}(n, p)$$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Large n approximation:

$$X \approx N(np, np(1 - p))$$

Probability involving percentage increase translates into inequality in X .

8. Conditional Probability with Exponential Variables

Using survival function:

$$P(X > t) = e^{-\lambda t}$$

Conditional probability:

$$P(X \geq t + s \mid X \geq t) = \frac{P(X \geq t + s)}{P(X \geq t)}$$

Apply exponential survival form.

9. Joint Transformation to Radial Variable

Let X and Y independent with pdfs $f_X(x)$, $f_Y(y)$.

Define:

$$Z = X^2 + Y^2$$

To obtain pdf:

Use joint density:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

Transform to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta$$

Jacobian:

$$J = r$$

Integrate over angle:

$$f_Z(r) = \int_0^{2\pi} f_X(r \cos \theta) f_Y(r \sin \theta) r \, d\theta$$

10. Order Statistics and Joint Distributions

Let $X_1, X_2, \dots$ be i.i.d. with CDF F .

Define:

$$M_n = \max(X_1, \dots, X_n)$$

10.1 Distribution of Maximum

$$P(M_n \leq x) = P(X_1 \leq x, \dots, X_n \leq x)$$

Using independence:

$$= [F(x)]^n$$

Thus:

$$F_{M_n}(x) = [F(x)]^n$$

10.2 Joint Distribution of M_n and M_{n+1}

Observe:

$$M_{n+1} = \max(M_n, X_{n+1})$$

Joint event:

$$P(M_n \leq x, M_{n+1} \leq y)$$

Case distinction:

If $y < x$: probability is zero.

If $y \geq x$:

$$P(M_n \leq x, X_{n+1} \leq y) = [F(x)]^n F(y)$$

This characterizes the joint CDF structure.

Summary of Core Concepts

This tutorial requires mastery of:

- Uniform distribution and geometric probability
- Normal distributions and mixture models
- Bayes' rule for posterior probabilities
- Minimization of expected absolute deviation (median property)
- Exponential distribution and memorylessness
- Mixture of exponentials
- Poisson distribution and conditional structure
- Binomial modeling of repeated independent trials
- Normal approximation to Binomial
- Transformation of random variables (CDF and Jacobian methods)
- Order statistics and maxima distributions